Problem 5.9

John borrows \$60 from Eddie. If he repays Eddie \$63 after 5 weeks, what simple interest has John paid? Use actual/actual for days counting. Assume a non-leap year.

The total amount repaid is \$63, so the interest paid is

$$I = 63 - 60 = 3.$$

We use the simple interest formula:

$$I = Prt,$$

where $P = 60$, $I = 3$, and t is the time in years.

Time

Since 5 weeks is

$$5 \cdot 7 = 35 \text{ days},$$

and the problem asks for actual/actual with a non-leap year, we use

$$t = \frac{35}{365}.$$

Solve for the rate

Substituting into $I = Prt$:

$$3 = 60 \cdot r \cdot \frac{35}{365}.$$

Solving for r,

$$r = \frac{3}{60} \cdot \frac{365}{35}.$$

Therefore,

$$r \approx 0.5214.$$

So the simple interest rate is

$$\boxed{52.14\%}.$$